

IEORE4004

Optimization Models and Methods

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Optimizing Investment Portfolios: Markowitz Mean-Variance Method

The Fundamental Investment Challenge

The Investor's Dilemma: Balancing Risk and Return

- Every investor wants to **maximize their returns**.
- Every investor wants to **minimize their risk** (avoiding large losses).
- These two goals are often in conflict.



HK	19300	19.3	8500/T	8642/T	8661/T	8730/T	8765/T	8783/T	8798/T	8823/T
150 TP	4055	-43.50	HK	HK	HK	HK	HK	HK	HK	HK
	2621	2571	1186	1462	1446	2514	1029	8227	2415	3427
	1143.89	1041	-13	+18	-21	+18	-97	-08	+74	-13
TKY	95.37	-181	2492/T	2514/T	2528/T	2534/T	2541/T	2553/T	2567/T	2572/T
240 TP	7865.20	-51.50	TKY	TKY	TKY	TKY	TKY	TKY	TKY	TKY
	142.30	136.89	2312	1651	1067	1929	27112	1721	1030	2469
	1991	-9.67	-09	-13	-32	-98	+65	-34	-87	+76
NY	37280	+1.89	4519/T	4542/T	4598/T	4602/T	4630/T	4698/T	4708/T	4751/T
95.7 TP	897.56	892.16	NY	NY	NY	NY	NY	NY	NY	NY
	351.79	326.51	1891	2019	1678	1254	1008	5761	1253	1875
	2312	-20.14	-25	-42	+16	-54	-12	-34	+46	+21
UK	31.25-29	29.45	1834/T	1865/T	1887/T	1899/T	1928/T	1945/T	1972/T	1991/T
225 TP	981.43	902.98	UK	UK	UK	UK	UK	UK	UK	UK
	103	-21.03	10879	10605	1762	2711	1933	1535	1111	2423
	5318	-11	-11	+07	-24	-17	-46	+24	-11	-64
CN	902	86	519/T	556/T	2589/T	26	2653/T	265	31	337/T
12.8 TP	21	19	CN	CN	CN	CN	CN	CN	CN	CN
	172	1987	1987	1987	1987	1991	34	6	741	741
	639	+19	+19	+19	+19	-39	-39	-39	-39	-39
RTS	2534	9716	4513/T	4513/T	4513/T	4513/T	4513/T	4513/T	4513/T	4513/T
850	59.31	59.31	RTS	RTS	RTS	RTS	RTS	RTS	RTS	RTS
	1943	-13	-13	-13	-13	-12	-12	-12	-12	-12

How do we systematically manage this trade-off?

The "Before Markowitz" Era: Intuition Over Precision

Diversification: A Good Idea, But How Much?

- **The Wisdom:** Investors intuitively knew "don't put all your eggs in one basket." Diversification was understood to be beneficial.
- **The Missing Piece:** However, there was no rigorous, quantitative way to:
 - **Measure** how different assets move together (their correlation).
 - **Quantify** the exact risk reduction gained by adding a specific asset.
 - **Construct a portfolio** that optimally balances risk and return.
- *Decisions were often based on gut feeling, industry diversification, or subjective rules of thumb.*

Enter Harry Markowitz: A Revolutionary Framework

● Who was Harry Markowitz?

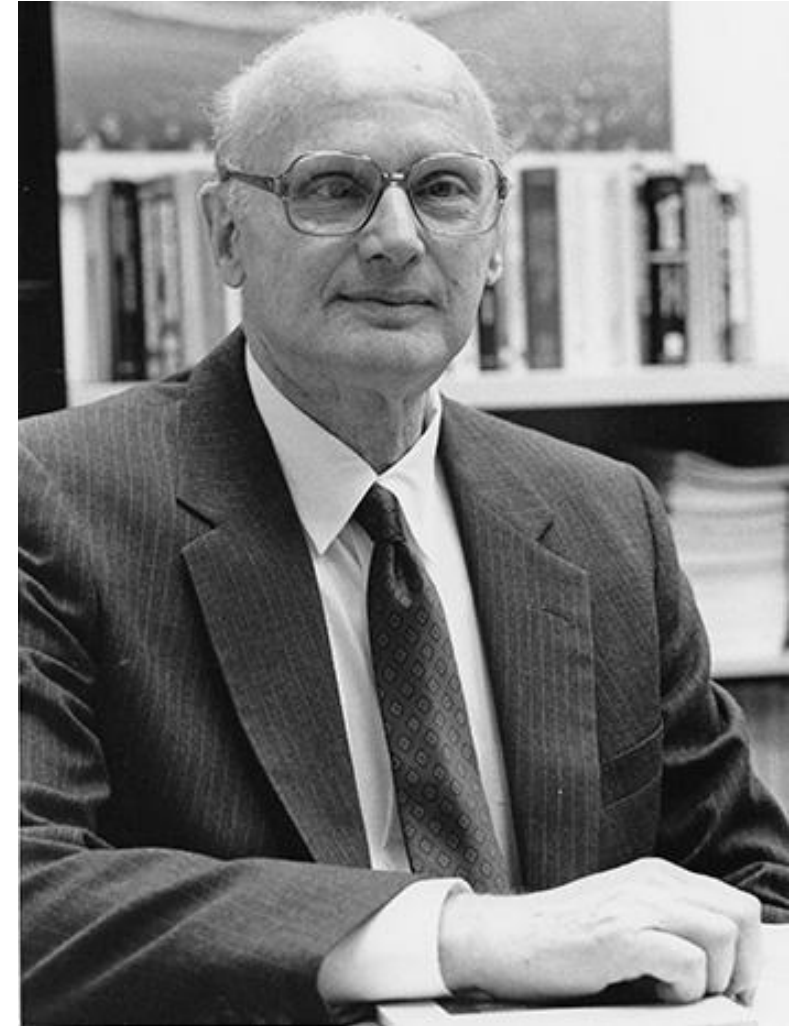
An economist whose groundbreaking work in the 1950s revolutionized how we think about investing.

He moved portfolio management from an art to a science.

His work on Modern Portfolio Theory earned him a Nobel Prize in Economics in 1990!

Markowitz, H. (1952). Portfolio Selection. *The Journal of Finance*, 7(1), 77–91. <https://doi.org/10.2307/2975974>

<https://www.youtube.com/watch?v=2pz1HJnD01c> (00:00 – 03:20)



Modern Portfolio Theory (MPT): A Scientific Approach

● What is MPT?

A mathematical framework for constructing portfolios that maximize expected return for a given level of risk or minimize risk for a given expected return.

Key Idea:

- *Focus on the portfolio as a whole!*
- *Consider not just individual asset performance, but also how assets move together (their covariance).*

The Core Concepts of MPT

Risk, Return, and the Power of Diversification

- **Risk and Return Trade-off:** Investors are generally risk-averse. They want higher returns but dislike higher volatility (risk).
- **Portfolio Metrics:** A portfolio's expected return and risk are determined by the expected returns, variances, and covariances of its constituent assets.
- **Covariance Matters:** Combining assets that don't move perfectly in sync can reduce overall portfolio risk. This is the essence of diversification's power.

Quantifying the Trade-off: The Mean-Variance Problem

Our Goal: Minimize Risk for a Target Return

● Let's Define Our Portfolio:

- We have N stocks.
- x_i : the proportion of our total investment in stock i .
- $\mathbf{x}$: the vector of all portfolio weights.
- μ_i : the expected annual return of stock i .
- $\boldsymbol{\mu}$: the vector of expected annual returns.
- Σ_{ij} : the covariance between stock i 's and stock j 's returns.
- $\boldsymbol{\Sigma}$: the covariance matrix.

Quantifying the Trade-off: The Mean-Variance Problem

The Math:

- Portfolio Expected Return: $R_p = \sum_{i=1}^N x_i \mu_i = \boldsymbol{\mu}^T \mathbf{x}$
- Portfolio Variance (Risk): $V_p = \sum_{i=1}^N \sum_{j=1}^N x_i x_j \Sigma_{ij} = \mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x}$
- **The Optimization Problem:**

Minimize $\mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x}$

Subject to $\boldsymbol{\mu}^T \mathbf{x} \geq \mu_{\text{target}}$ (Achieve at least our target return μ_{target})

$$\sum_{i=1}^N x_i = 1 \quad (\text{Invest all our capital})$$

$$x_i \geq 0, \quad \forall i = 1, \dots, N \quad (\text{No short-selling for simplicity})$$

Quick Note: What Is “Short Selling”?

1. Borrow the stock

You ask your broker to *lend* you shares of a company (e.g., 100 shares of A).

2. Sell the borrowed stock

You immediately sell those borrowed shares on the market; (e.g., at **\$50 each**, receiving **\$5,000**)

3. Wait for the price to fall

Your hope: the stock price drops. (e.g., if it falls to **\$30**, great!)

4. Buy back the stock at the new (lower) price

You now **buy 100 shares at \$30**, paying **\$3,000**.

5. Return the borrowed shares to the broker

You deliver the 100 shares back to your broker.

Your profit = Sell price – Buyback price (Here: Profit = \$5,000 – \$3,000 = **\$2,000**)

Recognizing the Structure: Quadratic Programming!

This Looks Familiar... It's Quadratic Programming!

Our Optimization Problem:

- Objective Function: $\mathbf{x}^T \Sigma \mathbf{x}$ (Quadratic in $\mathbf{x}$)
- Constraints: $\boldsymbol{\mu}^T \mathbf{x} \geq \mu_{\text{target}}$ and $\sum \mathbf{x}_i = 1$ (Linear)

Why QP is Perfect:

This is a classic Quadratic Programming (QP) problem. The ability to precisely define and solve these problems with tools like Gurobi is what makes MPT actionable – no more action based on gut feelings!

Our Example: 100 Established Stocks

Bringing it to Life with Real Data

Our Dataset:

- We've gathered historical price data for 100 established stocks across various sectors from April 2024 - April 2026).

Data Preparation:

- We downloaded adjusted closing prices (accounting for dividends and splits) using [yfinance](#).
- We calculated annualized expected returns (average historical return).
- We calculated the annualized covariance matrix (how returns moved together historically).

The Input:

This gives us our μ (expected returns) and Σ (covariance matrix) for the 100 stocks.

Our Example: 100 Established Stocks

Optimizing for a 25% Expected Annual Return

The Problem Setup:

Objective:

Minimize Portfolio Variance (Risk)

Constraints:

Expected Return $\geq 25\%$ (Our target)

Sum of Weights = 1 (Full investment)

Weights ≥ 0 (No short-selling)

Our Example: 100 Established Stocks

Optimizing for a 25% Expected Annual Return

Gurobi's Solution:

Status: Optimal Solution Found

Achieved Expected Return: 25.00% (Precisely our target!)

Minimum Portfolio Variance: 0.00924

Portfolio Volatility (Std Dev): $\approx 9.6\%$

Our Example: 100 Established Stocks

Optimizing for a 25% Expected Annual Return

Key Holdings:

- Let's look at the top holdings:
 - JNJ: 15.94%, DUK: 15.9%, KO: 15.5%, GOOGL: 0.0731%, XOM: 0.073, ...

What This Means:

- To achieve an expected annual return of 25% with the lowest possible risk (9.6% volatility), our historical data suggests allocating capital in these proportions to these 100 stocks.

Diversification:

- Notice how the investment is spread across different companies.

Our Example: 100 Established Stocks

Risk vs. Return:

We achieved our 25% target return. If we had aimed for a higher target return (e.g., 35%), the optimizer would likely have found a portfolio with higher risk (volatility) but also a higher expected return. This shows the fundamental risk-return trade-off.

Our Example: 100 Established Stocks

Try this yourself:

- Adjust the target return
 - Tryout some higher returns and lower ones (e.g., 35%, and 15%)
 - Observe how the portfolio variance changes (risk)
 - Observe how portfolio weight changes – for higher returns, your portfolio might be less diversified and focus on the individual stocks with higher historical returns – **Why might this be a problem?**

Visualizing the Efficient Frontier

